

Exercise J - A

1. The coefficients of three consecutive terms of $(1+x)^{n+5}$ are in the ratio 5 : 10 : 14. Then $n =$

[JEE(Adv.)2013]

$(1+x)^{n+5}$ के तीन क्रमागत पदों के गुणांक 5 : 10 : 14 के अनुपात में हैं। तब, $n =$

[JEE(Adv.)2013]

Ans. 6

Sol. Let the three consecutive terms be

$${}^{n+5}C_{r-1}, {}^{n+5}C_r, {}^{n+5}C_{r+1}$$

$$\therefore \frac{{}^{n+5}C_{r-1}}{{}^{n+5}C_r} = \frac{1}{2} \Rightarrow \frac{r}{n-r+6} = \frac{1}{2}$$

$$\Rightarrow n = 3r - 6 \quad \dots\dots(1)$$

$$\text{Also, } \frac{{}^{n+5}C_r}{{}^{n+5}C_{r+1}} = \frac{5}{7} \Rightarrow \frac{r+1}{n-r+5} = \frac{5}{7}$$

$$\Rightarrow 12r = 5n + 18 \quad \dots\dots(2)$$

Solving (1) and (2), we get $n = 6$

2. Coefficient of x^{11} in the expansion of $(1+x^2)^4(1+x^3)^7(1+x^4)^{12}$ is :

[JEE(Adv.)2014]

$(1+x^2)^4(1+x^3)^7(1+x^4)^{12}$ के प्रसार में x^{11} का गुणांक होगा

[JEE(Adv.)2014]

- (A) 1051 (B) 1106 (C) 1113 (D) 1120

Ans. C

Sol. Coefficient of $x^{11} \equiv \frac{(1+x^2)^4(1+x^3)^7(1+x^4)^{12}(1-x^2)^4}{(1-x^2)^4}$

$$\begin{aligned} \text{Coefficient of } x^{11} &\equiv (1-x^8)^4(1+x^4)^8(1+x^3)^7(1-x^2)^{-4} \\ &= (1-4x^8)(1+x^4)^8(7x^3+35x^9)(1-x^2)^{-4} \\ &= (7x^3+35x^9-28x^{11})(1+x^4)^8(1-x^2)^{-4} \end{aligned}$$

$$\begin{aligned} \text{Coefficient of } x^8 &= (7x+35x^6-28x^8)(1+8x^4+28x^8)(1-x^2)^{-4} \\ &= (7+35x^6-28x^8+56x^4+196x^8)(1-x^2)^{-4} \end{aligned}$$

$$\begin{aligned} \text{Coefficient of } t^4 &\equiv (7+56t^2+35t^3+168t^4)(1-t)^{-4} \\ &= 7 \cdot {}^7C_3 + 56 \cdot {}^5C_3 + 35 \cdot {}^4C_3 + 168 \\ &= 245 + 700 + 168 = 1113. \end{aligned}$$

Alterantive :

$$2x + 3y + 4z = 11$$

$$(x, y, z) = (0, 1, 2) \quad {}^4C_0 \times {}^7C_1 \times {}^{12}C_2$$

$$(1, 3, 0) \quad {}^4C_1 \times {}^7C_3$$

$$(2, 1, 1) \quad {}^4C_2 \times {}^7C_1 \times {}^{12}C_1$$

$$(4, 1, 0) \quad {}^7C_1$$

$$\text{coefficient of } x^{11} = 66 \times 7 + 35 \times 4 + 42 \times 12 + 7$$

$$= 1113$$

3. The coefficient of x^9 in the expansion of $(1+x)(1+x^2)(1+x^3)\dots(1+x^{100})$ is : [JEE(Adv.)2015]

$(1+x)(1+x^2)(1+x^3)\dots(1+x^{100})$ के विस्तार में x^9 के गुणांक का मान है। [JEE(Adv.)2015]

Ans. 8

Sol. $9 = (0, 9), (1, 8), (2, 7), (3, 6), (4, 5)$ # 5 cases

$9 = (1, 2, 6), (1, 3, 5), (2, 3, 4)$ # 3 cases

total = 8

4. Let m be the smallest positive integer such that the coefficient of x^2 in the expansion of $(1+x)^2 + (1+x)^3 + \dots + (1+x)^{49} + (1+mx)^{50}$ is $(3n+1) {}^{51}C_3$ for some positive integer n . Then the value of n is [JEE(Adv.)2016]

माना कि m ऐसा न्यूनतम धनात्मक पूर्णांक है कि $(1+x)^2 + (1+x)^3 + \dots + (1+x)^{49} + (1+mx)^{50}$ के विस्तार में x^2 का गुणांक $(3n+1) {}^{51}C_3$ किसी धनात्मक पूर्णांक n के लिए है। तब n का मान है। [JEE(Adv.)2016]

Ans. 5

Sol. ${}^2C_2 + {}^3C_2 + {}^4C_2 + \dots + {}^{49}C_2 + {}^{50}C_2 \cdot m^2 = (3n+1) {}^{51}C_3$

$$\Rightarrow {}^{50}C_3 + {}^{50}C_2 \cdot m^2 = (3n+1) {}^{51}C_3$$

$$\Rightarrow {}^{50}C_3 + {}^{50}C_2 + (m^2 - 1) {}^{50}C_2 = 3n {}^{51}C_3 + {}^{51}C_3$$

$$\Rightarrow n = \frac{m^2 - 1}{51}$$

$\Rightarrow m^2 = 51n + 1$ must be a perfect squared

$\Rightarrow n = 5$ and $m = 16$

5. Let

$$X = \binom{10}{1} C_1^2 + 2 \binom{10}{2} C_2^2 + 3 \binom{10}{3} C_3^2 + \dots + 10 \binom{10}{10} C_{10}^2,$$

where ${}^{10}C_r, r \in \{1, 2, \dots, 10\}$ denote binomial coefficients. Then, the value of $\frac{1}{1430} X$ is _____.

[JEE(Adv.)2018]

माना कि

$$X = \binom{10}{1}^2 + 2\binom{10}{2}^2 + 3\binom{10}{3}^2 + \dots + 10\binom{10}{10}^2,$$

जहाँ ${}^{10}C_r$, $r \in \{1, 2, \dots, 10\}$, द्विपद गुणांकों को दर्शाते हैं। तब $\frac{1}{1430} X$ का मान है।

[JEE(Adv.)2018]

Ans. 646

Sol.
$$X = \sum_{r=1}^{10} r \cdot {}^{10}C_r = 10 \cdot \sum_{r=1}^{10} {}^9C_{r-1} \cdot {}^{10}C_{10-r} = 10 \cdot {}^{19}C_9$$

$$\text{Now } \frac{X}{1430} = \frac{10 \cdot {}^{19}C_9}{1430} = \frac{{}^{19}C_9}{143} = \frac{{}^{19}C_9}{11 \times 13} = \frac{19 \cdot 17 \cdot 16}{8} = 19 \times 34 = 646$$

6. For nonnegative integers s and r , let
$$\binom{s}{r} = \begin{cases} \frac{s!}{r!(s-r)!} & \text{if } r \leq s, \\ 0 & \text{if } r > s. \end{cases}$$

For positive integers m and n , let
$$g(m, n) = \sum_{p=0}^{m+n} \frac{f(m, n, p)}{\binom{n+p}{p}}$$

where for any nonnegative integers p ,
$$f(m, n, p) = \sum_{i=0}^p \binom{m}{i} \binom{n+i}{p} \binom{p+n}{p-i}$$

Then which of the following statements is / are TRUE ?

[JEE(Adv.)2020]

- (A) $g(m, n) = g(n, m)$ for all positive integers m, n
- (B) $g(m, n+1) = g(m+1, n)$ for all positive integers m, n
- (C) $g(2m, 2n) = 2g(m, n)$ for all positive integers m, n
- (D) $g(2m, 2n) = (g(m, n))^2$ for all positive integers m, n

माना कि ऋणेतर् पूर्णाकों s एवं r के लिए $\binom{s}{r} = \begin{cases} \frac{s!}{r!(s-r)!} & \text{if } r \leq s, \\ 0 & \text{if } r > s. \end{cases}$

है। माना कि धनात्मक पूर्णाको m एवं n के लिए $g(m, n) = \sum_{p=0}^{m+n} \frac{f(m, n, p)}{\binom{n+p}{p}}$

है, जहाँ किसी ऋणेतर् पूर्णाक p के लिए $f(m, n, p) = \sum_{i=0}^p \binom{m}{i} \binom{n+i}{p} \binom{p+n}{p-i}$

है। तब निम्न में से कौन सा (से) कथन सही है (हैं) ?

[JEE(Adv.)2020]

- (A) सभी धनात्मक पूर्णाकों m, n के लिए $g(m, n) = g(n, m)$ है
- (B) सभी धनात्मक पूर्णाकों m, n के लिए $g(m, n+1) = g(m+1, n)$ है
- (C) सभी धनात्मक पूर्णाकों m, n के लिए $g(2m, 2n) = 2g(m, n)$ है
- (D) सभी धनात्मक पूर्णाकों m, n के लिए $g(2m, 2n) = (g(m, n))^2$ है

Ans. ABD

Sol.
$$f(m, n, p) = \sum_{i=0}^p \binom{m}{i} \frac{(n+i)!}{(n+i-p)!p!} \times \frac{(p+n)!}{(n+i)!(p-i)!}$$

$$= \sum_{i=0}^p \binom{m}{i} \left(\frac{n!}{(n+i-p)!(p-i)!} \right) \left(\frac{p+n!}{(n)!(p)!} \right)$$

$$= {}^{n+p}C_p \sum_{i=0}^p \binom{m}{i} {}^nC_{p-i}$$

$$f(m, n, p) = {}^{n+p}C_p \left(\binom{m}{0} {}^nC_p + \binom{m}{1} {}^nC_{p-1} + \dots + \binom{m}{p} {}^nC_0 \right)$$

$$= {}^{n+p}C_p \cdot {}^{m+n}C_p$$

$$g(m, n) = \sum_{p=0}^{m+n} \frac{{}^{n+p}C_p \cdot {}^{m+n}C_p}{{}^{n+p}C_p} = 2^{m+n}$$

(A) $g(m, n) =$

$(2)^{m+n} = (2)^{n+m} = g(n, m)$

$$(B) g(m, n+1) = (2)^{m+n+1} = (2)^{m+1+n}$$

$$(C) g(2m, 2n) = (2)^{2m+2n} = (2)^{2(m+n)}$$

$$(D) g(2m, 2n) = (2)^{2m+2n} = (2)^{2(m+n)}$$

$$= (2^{m+n})^2 = (g(m, n))^2$$

Hence, option (A), (B), (D) are correct answer.

7. Let a and b be two nonzero real numbers. If the coefficient of x^5 in the expansion of $\left(ax^2 + \frac{70}{27bx}\right)^4$ is equal to the coefficient of x^{-5} in the expansion of $\left(ax - \frac{1}{bx^2}\right)^7$, then the value of 2b is _____. [JEE(Adv.)2023]

माना कि a एवं b दो शून्येतर (nonzero) वास्तविक संख्याएं हैं। यदि $\left(ax^2 + \frac{70}{27bx}\right)^4$ के प्रसार (expansion) में x^5 का गुणांक (coefficient), $\left(ax - \frac{1}{bx^2}\right)^7$ के प्रसार में x^{-5} के गुणांक के बराबर है, तब 2b का मान है _____ [JEE(Adv.)2023]

Ans. 3

Sol. $T_{r+1} = {}^4C_r \left(ax^2\right)^{4-r} \cdot \left(\frac{70}{27bx}\right)^r$

$$= {}^4C_r \cdot a^{4-r} \cdot \frac{70^r}{(27b)^r} \cdot x^{8-3r}$$

$$\text{here } 8 - 3r = 5$$

$$8 - 5 = 3r \Rightarrow r = 1$$

$$\therefore \text{coeff.} = 4a^3 \cdot \frac{70}{27b}$$

$$T_{r+1} = {}^7C_r (ax)^{7-r} \left(\frac{-1}{bx^2}\right)^r$$

$$= {}^7C_r \cdot a^{7-r} \left(\frac{-1}{b}\right)^r \cdot x^{7-3r}$$

$$7 - 3r = -5 \Rightarrow 12 = 3r \Rightarrow r = 4$$

$$\text{Coeff.} : {}^7C_4 \cdot a^3 \cdot \left(\frac{-1}{b}\right)^4 = \frac{35a^3}{b^4}$$

$$\text{Now } \frac{35a^3}{b^4} = \frac{280a^3}{27b}$$

$$b^3 = \frac{35 \times 27}{280} = b = \frac{3}{2} \Rightarrow 2b = 3$$

CHALLENGER EXERCISES

Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. The coefficient of x^{53} in the expansion $\sum_{m=0}^{100} {}^{100}C_m (x-3)^{100-m} 2^m$ is :

$\sum_{m=0}^{100} {}^{100}C_m (x-3)^{100-m} 2^m$ के प्रसार में x^{53} का गुणांक होगा :

- (A) ${}^{100}C_{47}$ (B) ${}^{100}C_{53}$ (C) $-{}^{100}C_{53}$ (D) $-{}^{100}C_{100}$

Ans. C

Sol. $S = \sum_{m=0}^{100} {}^{100}C_m (x-3)^{100-m} 2^m$

$$S = {}^{100}C_0 (x-3)^{100} + {}^{100}C_1 (x-3)^{99} \cdot 2 + \dots + {}^{100}C_{100} \cdot 2^{100}$$

$$S = (2 + (x-3))^{100} = (x-1)^{100}$$

$$\text{Co-efficient of } x^{53} = -{}^{100}C_{53} = -{}^{100}C_{47}$$

2. If the sum of the coefficients in the expansion of $\left(x^2 + \frac{2}{x^3}\right)^n$ is 243, then the term independent of x , is equal to :

$\left(x^2 + \frac{2}{x^3}\right)^n$ के प्रसार में गुणांको का योगफल 243 है तब x से स्वतन्त्र पद होगा :

- (A) 11 (B) $\frac{7}{2}$ (C) 15 (D) 40

Ans. D

Sol. $\left(x^2 + \frac{2}{x^3}\right)^n = \sum_{r=0}^n {}^nC_r x^{2(n-r)} \left(\frac{2}{x^3}\right)^r$

$$= \sum_{r=0}^n {}^nC_r x^{2n-5r} \cdot 2^r$$

For independent term

$$2n = 5r$$

$$\Rightarrow r = 2$$

Given

$$\left(1 + \frac{2}{1}\right)^n = \sum_{r=0}^n {}^nC_r 2^r$$

$$= 243$$

$$\Rightarrow 3^n = 243$$

$$\Rightarrow n = 5$$

$\Rightarrow$ Coefficient

$$= {}^5C_2 \cdot 2^2 = 10 \cdot 4 = 40$$

3. The value of $4 \{ {}^nC_1 + 4 \cdot {}^nC_2 + 4^2 \cdot {}^nC_3 + \dots + 4^{n-1} \}$ is :

$4 \{ {}^nC_1 + 4 \cdot {}^nC_2 + 4^2 \cdot {}^nC_3 + \dots + 4^{n-1} \}$ का मान होगा :

- (A) 0 (B) $5^n + 1$ (C) 5^n (D) $5^n - 1$

Ans. D

Sol. $4 \{ {}^nC_1 + 4 \cdot {}^nC_2 + \dots + 4^{n-1} \cdot {}^nC_n \}$

$$= \sum_{r=1}^n {}^nC_r \cdot 4^r$$

$$= -{}^nC_0 + \sum_{r=0}^n {}^nC_r 4^r$$

$$= -1 + (1 + 4)^n$$

$$= 5^n - 1$$

4. The value of $\sum_{r=0}^n \frac{1}{(2r)!(2n-2r)!}$, is equal to :

$\sum_{r=0}^n \frac{1}{(2r)!(2n-2r)!}$, का मान होगा :

- (A) $\frac{2^{2n}}{(2n)!}$ (B) $\frac{2^{2n-1}}{(2n)!}$ (C) $\frac{2^{n-1}}{n!}$ (D) $\frac{2^n}{n!}$

Ans. B

Sol. $\sum_{r=0}^n \frac{1}{(2r)!(2n-2r)!} = \frac{1}{(2n)!} \sum_{r=0}^n \frac{(2n)!}{(2r)!(2n-2r)!}$

$$= \frac{1}{(2n)!} \sum_{r=0}^n {}^{2n}C_{2r}$$

$$= \frac{1}{(2n)!} [{}^{2n}C_0 + {}^{2n}C_2 + {}^{2n}C_4 + \dots + {}^{2n}C_{2n}]$$

$$= \frac{1}{\binom{2n}{2}} \cdot \frac{2^{2n}}{2} = \frac{2^{2n-1}}{\binom{2n}{2}}$$

5. In the expansion of $(1+x)^n (1+y)^n (1+z)^n$, the sum of the co-efficients of the terms of degree 'r' is :

; 1. गच्छ $(1+y)^n (1+z)^n$ के प्रसार में पदों की घात 'r' के रूप में गुणांको का योगफल होगा :

- (A) ${}^{n^3}C_r$ (B) ${}^nC_{r^3}$ (C) ${}^{3n}C_r$ (D) $3 \cdot {}^{2n}C_r$

Ans. C

Sol. Put $x=y=z$, we get $(1+x)^{3n}$

$\therefore$ coeff. of terms of degree r = ${}^{3n}C_r$

6. The sum of the last n coefficients in the expansion of $(1+x)^{2n-1}$ when expanding in ascending power of x, is equal to :

$(1+x)^{2n-1}$ के प्रसार में अन्त से n पदों के गुणांको का योगफल क्या होगा जबकि प्रसार x की बढ़ती हुई घात के रूप में किया जाए:

- (A) 2^{2n-1} (B) 2^{2n-2} (C) 2^{2n} (D) $2^{2n} - 2^n$

Ans. B

Sol. $(1+x)^{2n-1} = {}^{2n-1}C_0 + {}^{2n-1}C_1x + \dots + \underbrace{{}^{2n-1}C_nx^n + \dots + {}^{2n-1}C_{2n-1}x^{2n-1}}_{\text{last n terms}}$

$$\Rightarrow {}^{2n-1}C_n + {}^{2n-1}C_{n+1} + \dots + {}^{2n-1}C_{2n-1} = \frac{2^{2n-1}}{2} = 2^{2n-2} \quad \left\{ \because {}^nC_r = {}^nC_{n-r} \right\}$$

7. The fractional part of $\frac{3^{2n}}{8}$, is equal to :

- (A) $\frac{3}{8}$ (B) $\frac{7}{8}$ (C) $\frac{1}{8}$ (D) None of these

$\frac{3^{2n}}{8}$ का भिन्नात्मक भाग होगा :

- (A) $\frac{3}{8}$ (B) $\frac{7}{8}$ (C) $\frac{1}{8}$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\frac{3^{2n}}{8} = \frac{9^n}{8} = \frac{(8+1)^n}{8}$

$$= \frac{\sum_{r=0}^n {}^nC_r 8^r}{8} = \frac{{}^nC_0 + 8\lambda}{8}; \lambda \in I$$

$$= \frac{1}{8} + \lambda; \lambda \in I$$

Fractional part of $\frac{3^{2n}}{8}$ is $\frac{1}{8}$.

8. Given positive integers $r > 1$, $n > 2$, and the coefficients of the $(3r)^{\text{th}}$ and $(r + 2)^{\text{th}}$ terms in the binomial expansion of $(1 + x)^{2n}$ are equal, which, if any, of the following statements is true ?

दिया गया है धनात्मक पूर्णांक $r > 1$ के लिए $(r + x)^{2n}$ के प्रसार में $(3r)^{\text{th}}$ व $(r + 2)$ पदों के गुणांक बराबर है जहाँ $n > 2$ तब ?

- (A) $n = 2r$ (B) $n = 3r$ (C) $n = 2r + 1$ (D) None of these

Ans. A

Sol. $r > 1$, $n > 2$

$$(1 + x)^{2n} = \sum_{r=0}^{2n} {}^{2n}C_r x^r$$

Given

Coefficient of $3r^{\text{th}}$ term = Coefficient of $(r + 2)^{\text{th}}$ term

$$\Rightarrow {}^{2n}C_{3r-1} = {}^{2n}C_{r+1}$$

$$\Rightarrow 3r - 1 = r + 1 \quad \text{or} \quad 3r - 1 + r + 1 = 2n$$

$$\Rightarrow 2r = 2 \quad \text{or} \quad 4r = 2n$$

$$\Rightarrow r = 1 \quad \text{or} \quad n = 2r$$

as $r > 1$

$$\Rightarrow n = 2r$$

9. If in the expansion of $(1 + x)^m (1 - x)^n$, coefficients of x and x^2 are 3 and - 6 respectively, then m is :

; 1. ग द्वि^न $(1 - x)^n$ के प्रसार में x तथा x^2 के गुणांक 3 तथा - 6 है तब m बराबर होगा :

- (A) 6 (B) 9 (C) 12 (D) 24

Ans. C

Sol. Consider $(1+x)^m(1-x)^n$

$$\Rightarrow [{}^mC_0 + {}^mC_1x + {}^mC_2x^2 + \dots + {}^mC_mx^m] [{}^nC_0 + {}^nC_1(-x) + {}^nC_2(-x)^2 + \dots + {}^nC_n(-x)^n]$$

$$[\because (1+x)^n = {}^nC_0 + {}^nC_1x + \dots + {}^nC_nx^n]$$

On multiplying and comparing

$$\text{Now coefficient of } x = {}^mC_0 \cdot {}^nC_1(-1) + {}^mC_1 \cdot {}^nC_0 = m - n = 3 \dots (i)$$

$$\text{Coeff. of } x^2 = {}^mC_0 \cdot {}^nC_2(-1)^2 + {}^mC_2 \cdot {}^nC_0 + {}^mC_1 \cdot {}^nC_1(-1)$$

$$= \frac{n(n-1)}{2} + \frac{m(m-1)}{2} - mn = -6$$

$$= \frac{n(n-1)}{2} + \frac{(3+n)(3+n-1)}{2} - (3+n)n = -6 \quad \text{From (i)}$$

$$= \frac{n^2 - n}{2} + \frac{n^2 + 5n + 6}{2} - 3n - n^2 = -6$$

$$= \frac{n^2 - n + n^2 + 5n + 6 - 6n - 2n^2}{2} = -6$$

$$= -2n + 6 = -12$$

$$n! = 9 \quad \text{Now } m = 3 + n = 12$$

10. ${}^{n-1}C_r = (K^2 - 3) \cdot {}^nC_{r+1}$, if $K \in$

(A) $[-\sqrt{3}, \sqrt{3}]$ (B) $(-\infty, -2)$ (C) $(2, \infty)$ (D) $(\sqrt{3}, 2]$

Ans. D

Sol. ${}^{n-1}C_r = (k^2 - 3) \cdot {}^nC_{r+1}$

$$\frac{(n-1)!}{r!(n-r-1)!} = (k^2 - 3) \cdot \frac{n!}{(r+1)!(n-r-1)!}$$

$$\Rightarrow \frac{r+1}{n} + 3 = k^2$$

$$\Rightarrow \because 0 \leq r \leq n-1$$

$$1 \leq r+1 \leq n$$

$$\Rightarrow \frac{1}{n} \leq \frac{r+1}{n} \leq 1 \Rightarrow \frac{1}{n} \leq k^2 - 3 \leq 1$$

$$\Rightarrow 3 + \frac{1}{n} \leq k^2 \leq 4 \Rightarrow \sqrt{3 + \frac{1}{n}} \leq k \leq 2$$

$$\text{as } n \rightarrow \infty \Rightarrow \sqrt{3} < k \leq 2$$

$$\Rightarrow k \in (\sqrt{3}, 2]$$

11. The value of $\binom{30}{0}\binom{30}{10} - \binom{30}{1}\binom{30}{11} + \binom{30}{2}\binom{30}{12} - \dots + \binom{30}{20}\binom{30}{30}$ is, where $\binom{n}{r} = {}^nC_r$:

$\binom{30}{0}\binom{30}{10} - \binom{30}{1}\binom{30}{11} + \binom{30}{2}\binom{30}{12} - \dots + \binom{30}{20}\binom{30}{30}$ is, where $\binom{n}{r} = {}^nC_r$ का मान है:

- (A) $\binom{30}{10}$ (B) $\binom{30}{15}$ (C) $\binom{60}{30}$ (D) $\binom{31}{10}$

Ans. A

Sol. ${}^{30}C_0 {}^{30}C_{10} - {}^{30}C_1 {}^{30}C_{11} + {}^{30}C_2 {}^{30}C_{12} + \dots + {}^{30}C_{20} {}^{30}C_{30}$

= Coefficient of x^{20} in $(1 - x^2)^{30}$

as

$$(1 + x)^{30} = {}^{30}C_0 + {}^{30}C_1 x + {}^{30}C_2 x^2 + \dots + {}^{30}C_{20} x^{20} + \dots + {}^{30}C_{30} x^{30}$$

$$(1 - x)^{30} = {}^{30}C_0 - {}^{30}C_1 x + \dots + {}^{30}C_{30} x^{30}$$

$$= {}^{30}C_{30} - {}^{30}C_{29} x + {}^{30}C_{28} x^2 + \dots + {}^{30}C_0 x^{30}$$

$$= {}^{30}C_{30} - {}^{30}C_{29} x + {}^{30}C_{28} x^2 + \dots - {}^{30}C_{11} x^{19} + \dots + {}^{30}C_{30} x^{30}$$

$$= {}^{30}C_{10}$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

12. The polynomial $1 - x + x^2 - x^3 + \dots + x^{16} - x^{17}$ may be written in the form $\sum_{i=0}^{17} a_i y^i$ where $y = x + 1$ and a_i 's are constants. Then a_2 equals –

(A) The coefficient of t^3 in the expansion of $1 - (1 - t)^{18}$

(B) $\sum_{n=2}^{17} \binom{n}{2}$

(C) 816

(D) The coefficient of t^3 in the expansion of $1 - (1 - t)^{17}$

यदि बहुपद $1 - x + x^2 - x^3 + \dots + x^{16} - x^{17}$ को $\sum_{i=0}^{17} a_i y^i$ के रूप में लिखा जा सकता हो, जहाँ $y = x + 1$ तथा

a_i अचर है, तो a_2 बराबर होगा –

(A) $1 - (1 - t)^{18}$ के प्रसार में t^3 का गुणांक

(B) $\sum_{n=2}^{17} \binom{n}{2}$

(C) 816

(D) $1 - (1 - t)^{17}$ के प्रसार में t^3 का गुणांक

Ans. ABC

Sol. $1 - x + x^2 - x^3 + \dots + x^{16} - x^{17}$

$$= (1 - x) [1 + x^2 + \dots + x^{16}]$$

$$y = x + 1 \quad \frac{(1 - x)(1 - x^{18})}{(1 - x^2)}$$

$$\Rightarrow x = y - 1 \quad = \frac{1 - x^{18}}{(1 + x)}$$

$$= \frac{1 - (y - 1)^{18}}{y} = \frac{1 - \sum_{r=0}^{18} {}^{18}C_r y^r (-1)^r}{y}$$

$$= \frac{1 - [{}^{18}C_0 + {}^{18}C_1 y (-1) + {}^{18}C_2 y^2 (-1)^2 + \dots + {}^{18}C_{18} y^{18}]}{y}$$

$$= -\frac{1}{y} [-{}^{18}C_1 y + {}^{18}C_2 y^2 + \dots + {}^{18}C_{18} y^{18}]$$

$$= [{}^{18}C_1 - {}^{18}C_2 y + {}^{18}C_3 y^2 + \dots - {}^{18}C_{18} y^{17}]$$

$$a_2 = {}^{18}C_3 = \frac{18 \times 17 \times 16}{6} = 816$$

$$= \text{Coefficient of } t^3 \text{ in the expansion of } 1 - (1 - t)^{18}$$

$$= \sum_{n=2}^{17} {}^nC_2$$

13. In the expansion of $\left(x^3 + 3 \cdot 2^{-\log_{\sqrt{2}} \sqrt{x^3}}\right)^{11}$
- (A) there appears a term with the power x^2
- (B) there does not appear a term with the power x^2
- (C) there appears a term with the power x^{-3}
- (D) the ratio of the co-efficient of x^3 to that of x^{-3} is $1/3$

$$\left(x^3 + 3 \cdot 2^{-\log_{\sqrt{2}} \sqrt{x^3}}\right)^{11} \text{ के प्रसार में}$$

- (A) x^2 का पद आयेगा
- (B) x^2 का पद नहीं आयेगा
- (C) x^{-3} का पद आयेगा
- (D) x^3 एवं x^{-3} के गुणांको का अनुपात $1/3$ होगा।

Ans. BCD

Sol. $\left(x^3 + 3 \cdot 2^{-\log_{\sqrt{2}} \sqrt{x^3}}\right)^{11}$

$$\left\{ \log_{\sqrt{2}} \sqrt{x^3} = \frac{1}{2} \log_{\sqrt{2}} x^3 = \frac{3}{2} \log_{\sqrt{2}} x = 3 \log_2 x \right\}$$

$$= \left(x^3 + 3 \cdot 2^{\log_2 x^{-3}}\right)^{11}$$

$$= \left(x^3 + \frac{3}{x^3}\right)^{11}$$

$$= \sum_{r=0}^{11} {}^{11}C_r x^{33-3r} \cdot \frac{3^r}{x^{3r}}$$

$$= \sum_{r=0}^{11} {}^{11}C_r \cdot 3^r \cdot x^{33-6r}$$

For x^2 $33 - 6r = -3$
 $\Rightarrow 36 = 6r$

For x^3 $33 - 6r = 3$
 $\Rightarrow r = 5$

$$\frac{\text{Coeff. of } x^3}{\text{Coeff. of } x^{-3}} = \frac{{}^{11}C_5 \cdot 3^5}{{}^{11}C_6 \cdot 3^6} = \frac{6}{11-6+1} \cdot \frac{2}{3} = \frac{2}{6} = \frac{1}{3}$$

14. If it is known that the third term of the binomial expansion $\left(x + x^{\log_{10} x}\right)^5$ is 10^6 then x is equal to :

यदि $\left(x + x^{\log_{10} x}\right)^5$ के प्रसार में तीसरा पद 10^6 हो, तो x का मान होगा :

- (A) 10 (B) $10^{-5/2}$ (C) 100 (D) 5

Ans. AB

Sol. $\left(x + x^{\log_{10} x}\right)^5 = \sum_{r=0}^5 {}^5C_r x^{5-r} \left(x^{\log_{10} x}\right)^r$

Given

$${}^5C_2 x^3 \left(x^{\log_{10} x}\right)^2 = 10^6$$

$$\Rightarrow x^3 \left(x^{\log_{10} x}\right)^2 = 10^5$$

Take log both sides

$$3 \log_{10} x + 2 \log_{10} \left(x^{\log_{10} x}\right) = 5$$

$$\Rightarrow 3 \log_{10} x + 2 (\log_{10} x)^2 = 5$$

Let $\log_{10} x = t$

$$2t^2 + 3t - 5 = 0$$

$$\Rightarrow t = \frac{-3 \pm \sqrt{9+40}}{4}$$

$$t = \frac{-3 \pm 7}{4}$$

$$\Rightarrow t = 1 \quad \text{or} \quad t = -\frac{10}{4}$$

$$\Rightarrow x = 10 \quad \text{or} \quad \log_{10} x = -\frac{5}{2}$$

$$\Rightarrow x = 10 \quad \text{or} \quad x = 10^{-\frac{5}{2}}$$

15. The expansion of $\left(\sqrt{x} + \frac{1}{2\sqrt[4]{x}}\right)^n$ is arranged in decreasing power of x . If coefficient of first three terms form an A.P. then in expansion, the integral powers of x are :

$\left(\sqrt{x} + \frac{1}{2\sqrt[4]{x}}\right)^n$ के प्रसार को x की अवरोही घातों के क्रम में लिखा जाता है। यदि प्रथम पदों के गुणांक एक A.P. बनाते हों, तो प्रसार में, x की पूर्णांक घातें हैं :

- (A) 0 (B) 2 (C) 4 (D) 8

Ans. B

Sol.
$$\left(\sqrt{x} + \frac{1}{2\sqrt[4]{x}}\right)^n = \sum_{r=0}^n {}^nC_r (\sqrt{x})^{n-r} \cdot \frac{1}{2^r} \left(\frac{1}{x}\right)^{\frac{r}{4}}$$

$$= \sum_{r=0}^n {}^nC_r (x)^{\frac{n-r}{2}} \cdot \frac{1}{2^r} \cdot x^{-\frac{r}{4}}$$

$$= \sum_{r=0}^n {}^nC_r x^{\frac{2n-2r-r}{4}} \cdot \frac{1}{2^r}$$

$${}^nC_0 \cdot \frac{1}{2^0}, {}^nC_1 \cdot \frac{1}{2}, {}^nC_2 \cdot \frac{1}{2^2} \rightarrow \text{A.P.}$$

$$1 + \frac{n(n-2)}{8} = \frac{1}{2} \cdot n \cdot 2$$

$$\Rightarrow 8 + n^2 - n = 8n$$

$$\Rightarrow n^2 - 9n + 8 = 0$$

$$\Rightarrow n^2 - 8n - n + 8 = 0$$

$$\Rightarrow (n-1)(n-8) = 0$$

$$\Rightarrow n = 1 \text{ (wrong)} \quad \text{or} \quad n = 8$$

[For A.P. we need three terms in this question]

For integral power of x

$$\frac{2n-3r}{4} \in \mathbb{I}$$

$$\Rightarrow 2n - 3r \in 4\mathbb{I}$$

$$n = 8; r = 0, \dots, 7, 8$$

$$16 - 3r = 4k; k \in \mathbb{Z}$$

$$\Rightarrow r = \frac{16 - 4k}{3}$$

$$\Rightarrow r = 0, 8, 4$$

$\begin{matrix} \downarrow & \downarrow & \downarrow \\ 4 & -2 & 1 \end{matrix}$

16. $\sum_{i=0}^n \sum_{j=0}^n \sum_{k=0}^n \binom{n}{i} \binom{n}{j} \binom{n}{k}$, where $\binom{n}{r} = {}^nC_r$:

(A) is less than 5000 if $n = 3$

(B) is greater than 600 if $n = 3$

(C) is less than 5000 if $n = 4$

(D) is greater than 4000 if $n = 4$

$$\sum_{i=0}^n \sum_{j=0}^n \sum_{k=0}^n \binom{n}{i} \binom{n}{j} \binom{n}{k} \text{ जहाँ } \binom{n}{r} = {}^nC_r :$$

(A) 5000 से छोटा है यदि $n = 3$

(B) 600 से बड़ा है यदि $n = 3$

(C) 5000 से छोटा है यदि $n = 4$

(D) 4000 से बड़ा है यदि $n = 4$

Ans. ACD

Sol.
$$S = \sum_{i=0}^n \sum_{j=0}^n \sum_{k=0}^n \binom{n}{i} \binom{n}{j} \binom{n}{k}$$

$$= \sum_{i=0}^n \sum_{j=0}^n \left[{}^nC_i {}^nC_j \sum_{K=0}^n {}^nC_K \right]$$

$$= 2^n \sum_{i=0}^n {}^nC_i \left(\sum_{i=0}^n {}^nC_j \right)$$

$$= 2^n \cdot 2^n \cdot 2^n$$

$$= (2^n)^3$$